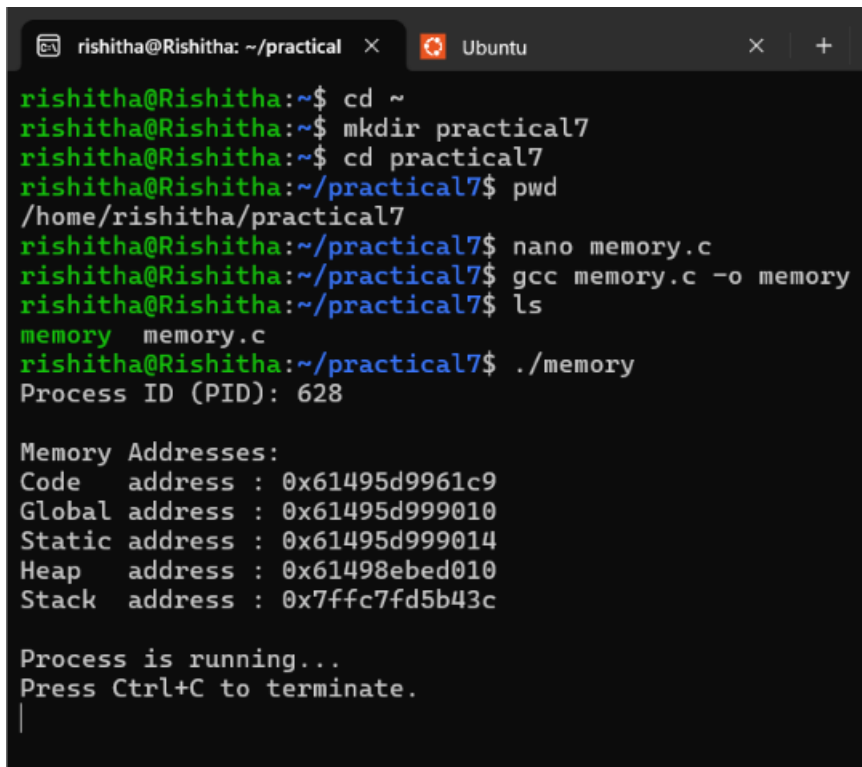


OS PRACTICAL

PRACTICAL-7

Write a program that prints the addresses of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment.

Make use of the `/proc//maps` file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings.

A terminal window titled 'rishitha@Rishitha: ~/practical' with an Ubuntu icon. The terminal shows a series of commands and their outputs. The user navigates to a directory named 'practical7', creates it if it doesn't exist, and then enters it. They compile a C program named 'memory.c' using 'gcc' and run it. The program outputs the addresses for Code, Global, Static, Heap, and Stack segments. The process ID (PID) is 628. The program also prints 'Process is running...' and 'Press Ctrl+C to terminate.' before the user presses Ctrl+C to exit.

```
rishitha@Rishitha:~$ cd ~
rishitha@Rishitha:~$ mkdir practical7
rishitha@Rishitha:~$ cd practical7
rishitha@Rishitha:~/practical7$ pwd
/home/rishitha/practical7
rishitha@Rishitha:~/practical7$ nano memory.c
rishitha@Rishitha:~/practical7$ gcc memory.c -o memory
rishitha@Rishitha:~/practical7$ ls
memory  memory.c
rishitha@Rishitha:~/practical7$ ./memory
Process ID (PID): 628

Memory Addresses:
Code   address : 0x61495d9961c9
Global address : 0x61495d999010
Static address : 0x61495d999014
Heap   address : 0x61498ebed010
Stack  address : 0x7ffc7fd5b43c

Process is running...
Press Ctrl+C to terminate.
|
```

```
rishitha@Rishitha: ~/practical7 × Ubuntu × + ▾ - □ ×
rishitha@Rishitha:~$ cat /proc/628/maps
61495d995000-61495d996000 r--p 00000000 08:30 39588 /ho
me/rishitha/practical7/memory
61495d996000-61495d997000 r-xp 00001000 08:30 39588 /ho
me/rishitha/practical7/memory
61495d997000-61495d998000 r--p 00002000 08:30 39588 /ho
me/rishitha/practical7/memory
61495d998000-61495d999000 r--p 00002000 08:30 39588 /ho
me/rishitha/practical7/memory
61495d999000-61495d99a000 rw-p 00003000 08:30 39588 /ho
me/rishitha/practical7/memory
61498ebd000-61498ec0e000 rw-p 00000000 00:00 0 [he
ap]
7d54e9600000-7d54e9628000 r--p 00000000 08:30 13647 /us
r/lib/x86_64-linux-gnu/libc.so.6
7d54e9628000-7d54e97c0000 r-xp 00028000 08:30 13647 /us
r/lib/x86_64-linux-gnu/libc.so.6
7d54e97c0000-7d54e980e000 r--p 001c0000 08:30 13647 /us
r/lib/x86_64-linux-gnu/libc.so.6
7d54e980e000-7d54e9812000 r--p 0020d000 08:30 13647 /us
r/lib/x86_64-linux-gnu/libc.so.6
7d54e9812000-7d54e9814000 rw-p 00211000 08:30 13647 /us
r/lib/x86_64-linux-gnu/libc.so.6
7d54e9814000-7d54e9821000 rw-p 00000000 00:00 0
7d54e982e000-7d54e9831000 rw-p 00000000 00:00 0
7d54e9838000-7d54e983a000 rw-p 00000000 00:00 0
7d54e983a000-7d54e983e000 r--p 00000000 00:00 0 [vv
ar]
7d54e983e000-7d54e9840000 r--p 00000000 00:00 0 [vv
ar_vclock]
7d54e9840000-7d54e9842000 r-xp 00000000 00:00 0 [vd
so]
7d54e9842000-7d54e9843000 r--p 00000000 08:30 13644 /us
r/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7d54e9843000-7d54e9872000 r-xp 00001000 08:30 13644 /us
r/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7d54e9872000-7d54e987d000 r--p 00030000 08:30 13644 /us
r/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7d54e987d000-7d54e987f000 r--p 0003b000 08:30 13644 /us
```

```
rishitha@Rishitha: ~/practical7 x Ubuntu
r/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7d54e9880000-7d54e9881000 rw-p 00000000 00:00 0
7ffc7fd3d000-7ffc7fd5e000 rw-p 00000000 00:00 0 [st
ack]
rishitha@Rishitha:~$ cat /proc/628/maps | grep -E '\[heap\]|\[stack\]|practi
cal7/memory'
61495d995000-61495d996000 r--p 00000000 08:30 39588 /ho
me/rishitha/practical7/memory
61495d996000-61495d997000 r-xp 00001000 08:30 39588 /ho
me/rishitha/practical7/memory
61495d997000-61495d998000 r--p 00002000 08:30 39588 /ho
me/rishitha/practical7/memory
61495d998000-61495d999000 r--p 00002000 08:30 39588 /ho
me/rishitha/practical7/memory
61495d999000-61495d99a000 rw-p 00003000 08:30 39588 /ho
me/rishitha/practical7/memory
61498ebed000-61498ec0e000 rw-p 00000000 00:00 0 [he
ap]
7ffc7fd3d000-7ffc7fd5e000 rw-p 00000000 00:00 0 [st
ack]
rishitha@Rishitha:~$ pmap -x 628
628: ./memory
Address Kbytes RSS Dirty Mode Mapping
000061495d995000 4 4 0 r---- memory
000061495d996000 4 4 0 r-x-- memory
000061495d997000 4 4 0 r---- memory
000061495d998000 4 4 4 r---- memory
000061495d999000 4 4 4 rw--- memory
000061498ebed000 132 4 4 rw--- [ anon ]
00007d54e9600000 160 156 0 r---- libc.so.6
00007d54e9628000 1632 1184 0 r-x-- libc.so.6
00007d54e97c0000 312 184 0 r---- libc.so.6
00007d54e980e000 16 16 16 r---- libc.so.6
00007d54e9812000 8 8 8 rw--- libc.so.6
00007d54e9814000 52 20 20 rw--- [ anon ]
00007d54e982e000 12 8 8 rw--- [ anon ]
00007d54e9838000 8 4 4 rw--- [ anon ]
00007d54e983a000 16 0 0 r---- [ anon ]
00007d54e983e000 8 0 0 r---- [ anon ]
```

```

00007d54e9840000      8      8      0 r-x--  [ anon ]
00007d54e9842000      4      4      0 r---- ld-linux-x86-64.so.2
00007d54e9843000    188    184      0 r-x-- ld-linux-x86-64.so.2
00007d54e9872000     44     44      0 r---- ld-linux-x86-64.so.2
00007d54e987d000      8      8      8 r---- ld-linux-x86-64.so.2
00007d54e987f000      4      4      4 rw--- ld-linux-x86-64.so.2
00007d54e9880000      4      4      4 rw--- [ anon ]
00007ffc7fd3d000    132     16     16 rw--- [ stack ]
-----
total kB          2768    1876    100
rishitha@Rishitha:~$ cat /proc/628/status | grep -E 'VmSize|VmRSS|VmData|VmStk|VmExe|VmLib'
VmSize:    2768 kB
VmRSS:    1876 kB
VmData:    224 kB
VmStk:    132 kB
VmExe:      4 kB
VmLib:    1828 kB
rishitha@Rishitha:~$

```